

THE 50 DOLLAR BATCH HEATER: FREE HOT WATER FROM WINTER SUN

A scrapped water heater tank, painted black and boxed behind an old window. The parts, the sun numbers for your region, and the freeze rules.

The box costs about 50 dollars when the tank and the glass are salvage. On a clear winter day across the southern half of the country, it lifts 20 gallons of cold water past 100 degrees. That's a long shower, and the sun sends no bill.

THE TRAP

85 to 90%	Of sunlight that ordinary window glass lets through.
About 96%	Of the light landing on it that flat black paint absorbs, turning it into heat in the steel, then the water.
About 10 microns	The infrared a 100°F tank gives off. Window glass blocks almost everything past about 5 microns, so the heat can't get back out.
About 3/4	The punch of a December noon sun versus June, when the glass faces it straight on.

- ☐ Behind glass, the air around the tank sits still, and still air is a poor conductor. That's most of why a parked car cooks.
- ☐ **Tilt:** about your latitude **plus 15 degrees**, facing south. Mid country, that's close to 55 degrees. A flat lid catches less than half the winter sun.
- ☐ **Stratification:** warm water rises to the top on its own. Draw from the top, feed cold in at the bottom, and you get the hottest water first, no pump.
- ☐ Why not a black hose coil? A half inch hose holds about 1 gallon per 100 feet. Hot fast, gone after one rinse.



Black tank, storm window front, foil lid propped open, tilted steep toward the low sun.

PARTS AND ROUGH COSTS

Item	Cost
Scrapped water heater tank (20 gallon lowboy, mobile home, or point of use unit is best)	Salvage
Storm window, 3 by 4 feet	Salvage
Flat black high heat grill paint	Under 8 dollars a can
Foil faced polyiso foam board, 2 sheets	About 20 dollars each
New temperature and pressure relief valve	About 20 dollars
New anode rod	About 20 to 30 dollars
Scrap plywood or pallet boards, silicone, brass drain valve, threaded plugs, hot water rated hose	Scrap and fittings
Box, salvaged tank and window	About 50 dollars
Valve, anode, fittings	Another 50 to 70
All new tank and glazing instead	Closer to 300

Winter upgrade: a sheet of clear twin wall polycarbonate behind the glass, plus foam board glued inside the lid.

HOW MUCH HEAT IT BANKS

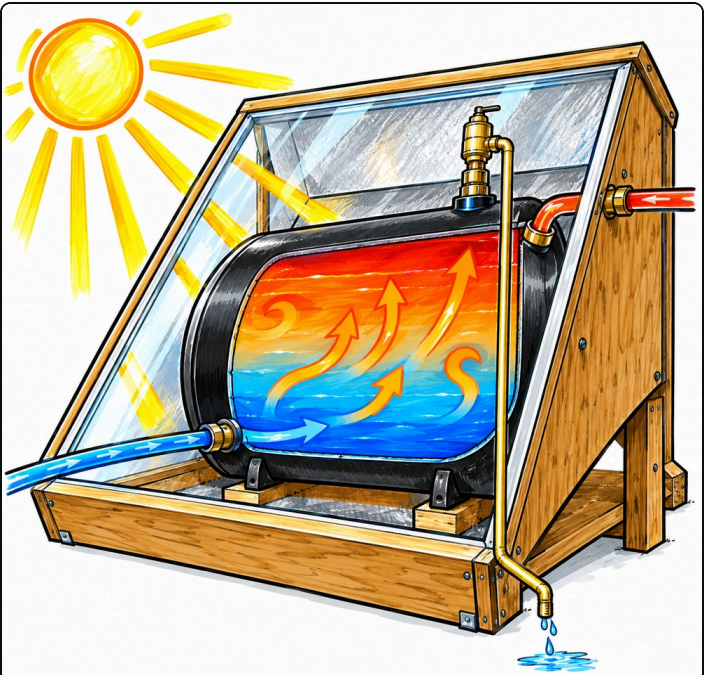
1 BTU lifts 1 pound of water by 1°F. 1 gallon is about 8 1/3 pounds.

167 lb	Water in a 20 gallon tank.
About 8,300 BTU	To lift it 50 degrees.
About 1,100	BTU per square foot per day, Kansas City in December, on a south face tilted for winter, cloudy days included.
About 13,700	BTU landing on a 3 by 4 foot (12 square foot) storm window on an average December day.
About half	Reaches the water, about 6,800 BTU, after reflection, the round tank, and the single pane leaking. A rise of about 40 degrees.

WHAT THAT MEANS WHERE YOU LIVE

City, December	Sun per sq ft per day	Average day result
Kansas City	About 1,100 BTU	Tap water in the low 40s finishes in the low to mid 80s. A clear day clears 100.
Phoenix	About 1,700 BTU	Around 110°F, clear and cloudy days together.
Seattle	Under 600 BTU	After losses, barely warms the water.

The honest map: the Department of Energy says batch heaters belong where it rarely freezes. Across the southern third of the country and the mild coasts, this box runs all winter. Farther north, it's a three season heater that also catches clear winter days, and you drain it before hard freezes.



Cold in at the lowest point, hot out at the highest, relief pipe aimed at the ground.

SIZE THE TANK RIGHT

- ☐ The sun delivers the same heat to 20 gallons or 40. Spread over 40 gallons, the rise is half as big and the water finishes in the 60s.
- ☐ Most tanks at the curb are 40 or 50 gallons. Hunt for a 20 gallon lowboy, a mobile home heater, or a point of use unit.
- ☐ Stuck with 40 gallons? Give it two storm windows side by side.

WHAT IT'S WORTH

- ☐ Water heating is about 18 percent of a typical home's energy, about 400 to 600 dollars a year.
- ☐ A commercial solar water heater, installed: about 3,000 to 9,000 dollars. The 30 percent federal tax credit ended with 2025.
- ☐ **Best as a preheater:** your heater normally lifts cold water 70 or 80 degrees. Feed it 85 degree water and the lift drops to about 50.
- ☐ On an electric heater, 20 gallons of preheat a day saves roughly 50 to 150 dollars a year, depending on your sun. In sunny regions it can pay for itself inside a year or two.
- ☐ **Blackout:** an electric tank stays warm maybe a day. A gas heater with electronic ignition stops too. The black tank keeps making hot water every clear day. You still need water pressure to fill it, and on a well, no power means no pump.

THE BUILD

- 1** **Test the tank first.** Fill it from a hose and let it stand a day. If nothing weeps, you've got a collector.
- 2** Cut off the outer jacket with snips. Scrape off the sprayed on foam. Slow, messy work.
- 3** Leave the heating elements in, or plug their ports with threaded plugs, so it holds water.
- 4** Flush out the sediment. Fit a new anode rod, or bare steel corrodes fast once the water warms.
- 5** Fit a **new** relief valve. Run a pipe from it out the bottom of the box, pointed at the ground.
- 6** Paint the tank flat black with high heat grill paint.
- 7** Build the box from scrap plywood or pallet boards, a couple of inches of clearance all around.
- 8** Line it with foil faced polyiso, foil side in. It handles the heat better than pink or blue foam.
- 9** Set the storm window in the open face, seal the edges with silicone. For winter, add twin wall polycarbonate behind it. It cuts the heat leaking back out roughly in half.
- 10** Hot outlet at the **highest** point, cold inlet at the **lowest**. Swap the plastic drain valve for brass.
- 11** Tilt the box on a frame at latitude plus 15, facing south.
- 12** On the ground or a stout low stand, **never on a roof**. The water alone is 167 pounds, and a plumbed box weighs well over 200.
- 13** Line the lid with foil to bounce extra sun in, and glue foam board to its inner face.
- 14** **Outdoor setup:** garden hose into the bottom, hot out the top to a shower head or laundry tub, hose rated for hot water.
- 15** **Into the house?** That's a plumber's job: rated pipe, a shutoff on each side of the box, and a bypass pipe between them.



Hose in at the bottom, hot out the top. Wrist test before anyone steps under it.

THE DAILY ROUTINE

About a minute, twice a day.

- ☐ Open the lid when the sun clears the trees.
- ☐ Draw hot water in the late afternoon, when the tank peaks.
- ☐ Refill with cold right after you draw, so that water has all of the next day to climb.
- ☐ Close the lid at dusk.

SAFETY RULES. NOT OPTIONAL.

- ☐ **Scald:** in Phoenix the tank finishes around 110 on an average December day, and a summer day pushes it well past 130. Water at 130°F burns adult skin in about 30 seconds, a child's even faster. **Test on your wrist before it touches a kid.**
- ☐ **Relief valve is mandatory.** Under hose pressure the tank is sealed. Never reuse the old valve; it can stick shut. If it opens, it sprays scalding water.
- ☐ **The lukewarm zone:** Legionnaires' bacteria grow between 77 and 113°F. Above about 120 they stop growing but survive. At 140 they die in about half an hour.
- ☐ Box feeds a house heater? Set that heater at **140**, with a mixing valve at its outlet set at **120**.
- ☐ Standalone? Use the water the same afternoon. Not using it for a few days? Drain it. Never breathe week old lukewarm water as shower mist.
- ☐ **Freeze:** the full tank in its box, foam lid shut, can take up to three days to freeze at 12°F, but the half inch pipes can freeze in one night. Short, insulated pipe runs. Close the lid every night. Hard freeze forecast? Drain the tank and pipes that day.
- ☐ **Don't drink from a standalone box.** Salvaged tank, unknown history, garden hose. Showers, dishes, and laundry only.

NEXT STEP

The same low winter sun hits your south wall every clear day, with nothing there to catch it. Aim the same trap at the air in your house and it heats a room: watch the 100 dollar sun wall video next.

Watch the full video: youtube.com/@SketchySurvival101

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